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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech. (Machine Design) Degree Examinations

May - June 2024

22MMD231 – COMPOSITE MATERIALS TECHNOLOGY

Duration: 3 Hours

Max. Marks:100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

❖ Missing data if any may be suitably assumed.

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Give the characteristics of composite materials. Also give the classification of composites based on matrix and reinforcement	8	L*1	1	1
b)	Discuss various fibers used in the composite manufacturing also explain the manufacturing of carbon fiber.	8	L2	1	1
c)	What are 'whiskers'? Name some whiskers that are currently available?	4	L1	1	1
2. a)	Derive the rule of mixtures for transverse modules of a unidirectional lamina using the strength of the material approach. State the assumptions to be made in the approach.	8	L2	2	2
b)	Derive the equation for elastic modulus of a composite laminate.	12	L2	2	2
3. a)	Write the number of independent elastic constants for three-dimensional anisotropic, monoclinic, orthotropic, transversely isotropic, and isotropic materials.	10	L2	2	2
b)	Find the following for a 60° angle lamina of graphite/epoxy whose properties are given by $E_1=180\text{GPa}$, $E_2=10\text{ GPa}$, $\nu_{12}=0.28$, $G_{12}=7\text{ GPa}$. (i) The transformed compliance matrix (ii) The transformed reduced stiffness matrix	10	L3	2	2
	Unit – II				
4. a)	Find the maximum value of $S>0$ if a stress $\sigma_x=2S$, $\sigma_y=-3S$, and $\tau_{xy}=4S$ are applied to a 60° lamina of graphite/epoxy. Use maximum stress theory and strain theory. Use the properties of a unidirectional graphite/epoxy lamina ($E_1=181\text{GPa}$, $E_2=10.3\text{GPa}$, $\nu_{12}=0.28$, $G_{12}=7.17\text{GP}$, $(s_2^C)_{ult} = 246\text{MPa}$, $t_{12} = 40\text{MPa}$).	10	L3	3	2
b)	Write a note on the Tsai-Wu Failure theory.	10	L2	3	2
5. a)	Explain the pultrusion process with a neat sketch. List its merits and demerits.	10	L2	4	1
b)	Explain the blow moulding with a neat sketch. List its merits and demerits.	10	L2	4	1
6. a)	What are the NDT methods used in the testing of composite materials? Explain any ultrasonic technique.	10	L2	4	1
b)	Explain laser beam cutting of composites, with a neat sketch.	10	L2	4	1
	Unit – III				
7. a)	Discuss the applications of polymer matrix composites in i) Sports ii) Space Hardware	12	L2	5	1
b)	Discuss advanced composites.	8	L2	5	1
8. a)	Explain the different types of MMCs.	10	L3	5	1
b)	Explain diffusion bonding and stir casting methods of producing MMCs.	10	L3	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
